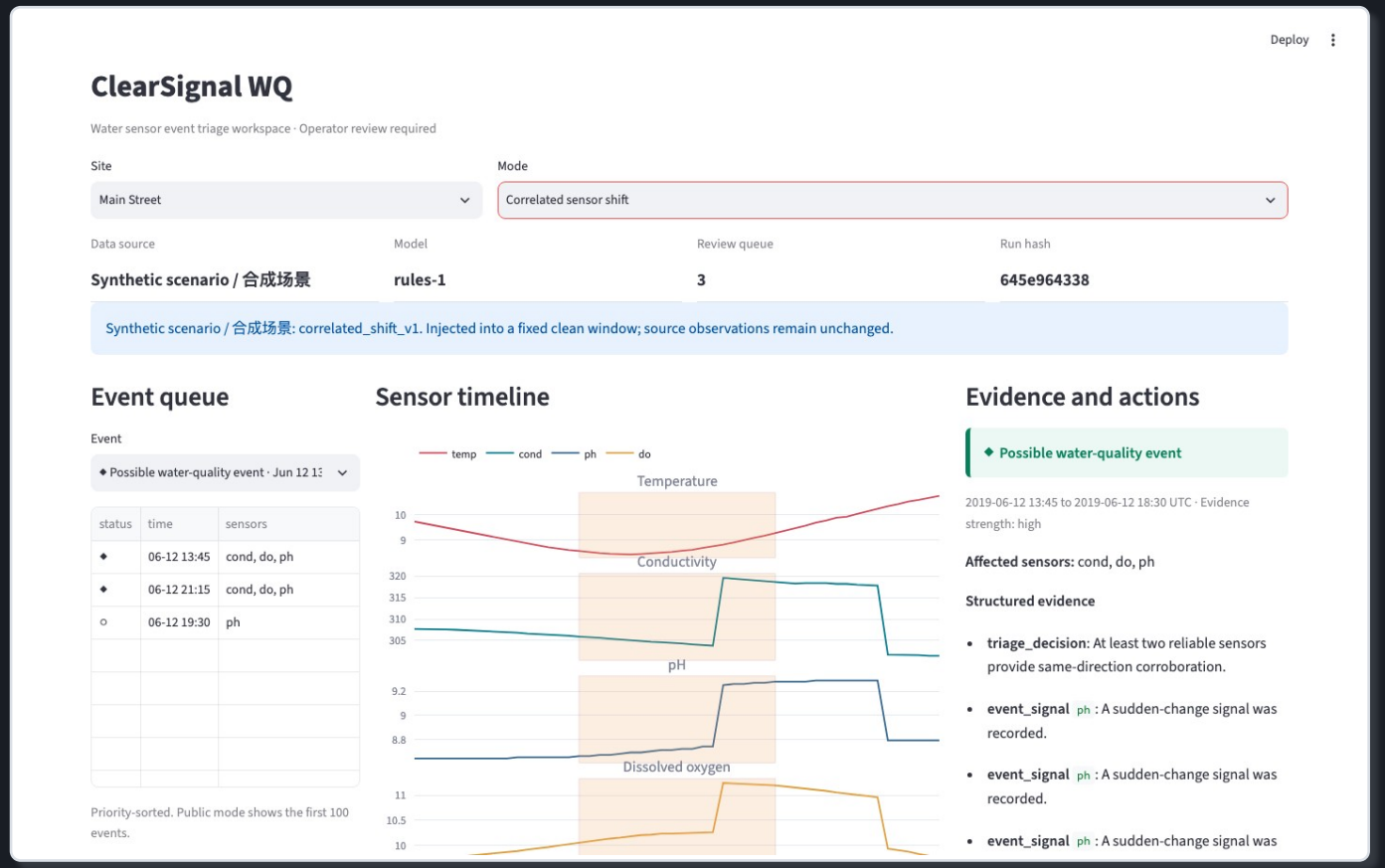
CLEARSIGNAL WQ

Evidence before escalation

Explainable AI triage for water sensor events, drift, and data-integrity failures.

OPERATOR REVIEW REQUIRED



Four outcomes. One review queue. Every decision keeps its evidence and next human action.

Qizhong Deng | Xylem Global Student Innovation Challenge 2026

THE PROBLEM

One unusual signal can imply three different actions

Small monitoring teams need a defensible next step, not another unexplained alert.

Environmental change

Several reliable sensors move together. The operator should retest in the field and decide whether laboratory confirmation is justified.

Sensor drift

A single sensor changes while reliability falls. The right response is maintenance, calibration, and an independent measurement.

Data gap

Timestamps or telemetry fail. The operator should inspect the data link before treating missing information as a water event.

The cost of confusing them alert fatigue | avoidable field work | delayed maintenance | inconsistent escalation

HOW IT WORKS

One deterministic path from readings to review

01

Validate

timestamps, sensors, provenance

02

Detect

gaps, range, stuck, robust jumps

03

Group

stable event windows and IDs

04

Assess

reliability and corroboration

05

Triage

fixed priority and explanation

06

Act

retest, calibrate, inspect, escalate

Why deterministic?

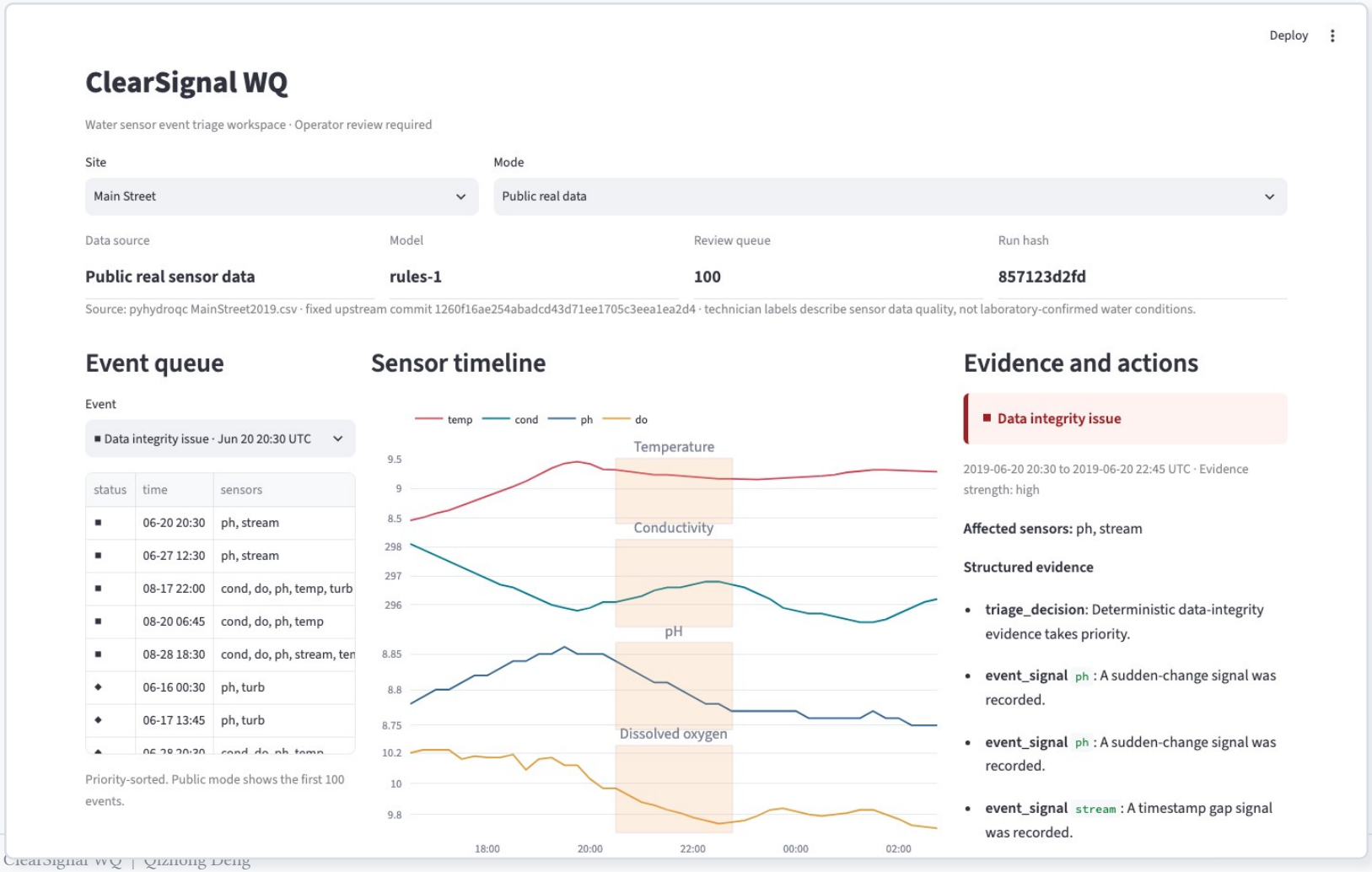
The same evidence always produces the same category, explanation, limitation, and human verification action.

The UI never recalculates thresholds or changes triage decisions.

THE PRODUCT

A review workspace built for repeated operational decisions

Queue the event, inspect the signal context, and record the next verification action.



1. Prioritize

Stable event IDs and severity-aware ordering.

2. Explain

Four-sensor timeline plus structured evidence.

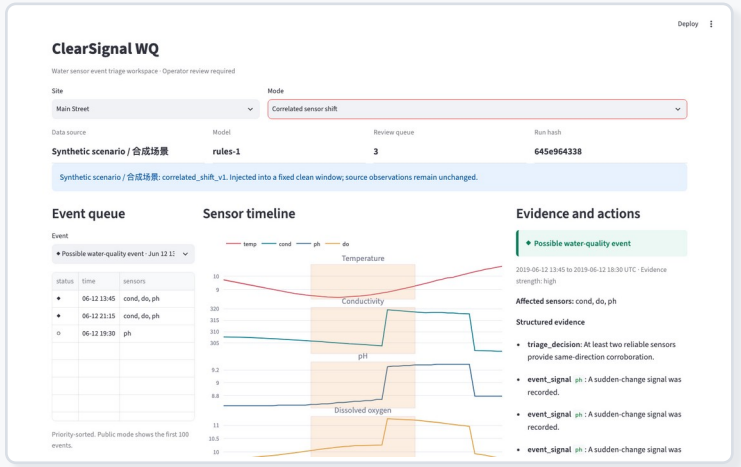
3. Verify

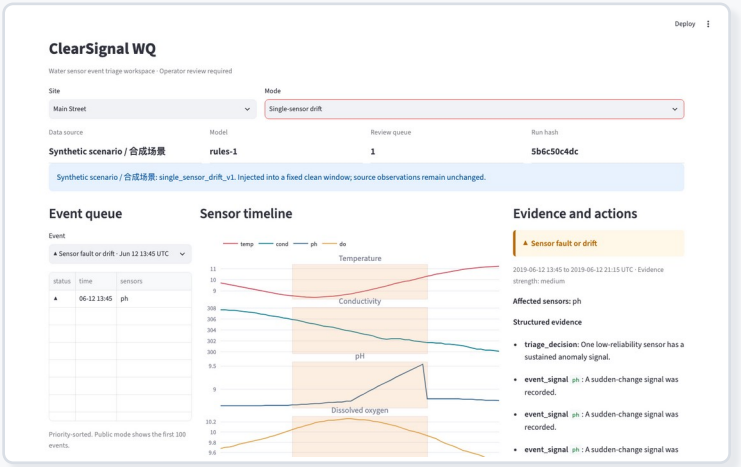
Retest, calibrate, inspect the link, or escalate.

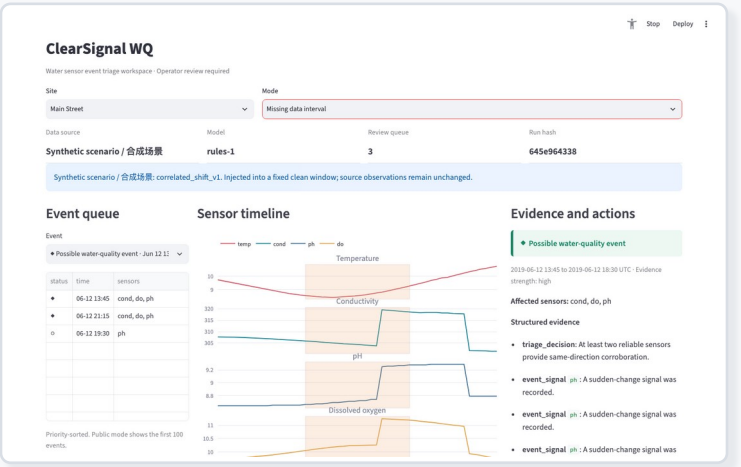
SCENARIO PROOF

The workflow changes when the evidence changes

Synthetic scenarios are explicitly labelled and test product behavior, not real-world accuracy.







Possible water-quality event

Multiple reliable sensors move together

Sensor fault or drift

One sensor changes while reliability falls

Data-integrity issue

Missing timestamps take priority

EVALUATION

We kept the simpler model because the evidence said so

Validation on technician-labelled sensor data-quality events

Model and split	Recall	Precision	Invalid/day	Decision
Rules / validation	1.000	0.0135	7.30	**Selected**
ARIMA / validation	0.800	0.0111	7.14	**Rejected**
Rules / frozen test	0.7368	0.0426	8.2895	**Held out**

7.30

invalid alerts / day
on validation

93.50%

package test
coverage

The limitation is part of the result

Low precision and high alert burden make this unsuitable for unattended deployment.

311 automated tests | 3 browser flows | 3 target viewports

RESPONSIBLE PATH

Decision support today. Field evidence next.

The prototype organizes evidence; it does not certify water safety or replace independent testing.

What it does

- Prioritizes sensor events
- Preserves evidence and limitations
- Makes the next human action explicit
- Keeps synthetic scenarios clearly labelled

What it does not do

- Confirm contamination
- Declare water safe
- Replace field or laboratory tests
- Operate unattended at the current alert rate

Next field pilot

- Add maintenance and calibration records
- Pair alerts with independent samples
- Measure avoidable escalations and response time
- Tune only on pilot evidence

github.com/Cedric-uxb/clearsignal-wq

Public code, data provenance, evaluation, and reproducible tests